



CERTIFICATE OF ACCREDITATION

The ANSI National Accreditation Board

Hereby attests that

Transcat - Toronto
1435 Norjohn Ct. #8-9
Burlington, ON L7L 0E6
Canada

Fulfills the requirements of

ISO/IEC 17025:2017

and national standards

ANSI/NCSL Z540-1-1994 (R2002) and
ANSI/NCSL Z540.3-2006 (R2013)

In the fields of

CALIBRATION AND DIMENSIONAL MEASUREMENT

This certificate is valid only when accompanied by a current scope of accreditation document.
The current scope of accreditation can be verified at www.anab.org.

R. Douglas Leonard Jr., VP, PILR SBU

Expiry Date: 07 September 2023
Certificate Number: AC-2489.23



This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017.
This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory
quality management system (refer to joint ISO-ILAC-IAF Communiqué dated April 2017).

SCOPE OF ACCREDITATION TO ISO/IEC 17025:2017

AND

ANSI/NCSL Z540-1-1994 (R2002)

ANSI/NCSL Z540.3-2006 (R2013)

Transcat-Toronto

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CALIBRATION AND DIMENSIONAL MEASUREMENT

Valid to: September 7, 2023

Certificate Number: AC-2489.23

CALIBRATION

Acoustics and Vibration

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Sound Level – Source ¹	125 Hz to 4 kHz		General Radio 1986 Sound Level Calibrator
	74 dB	0.73 dB	
	84 dB	0.62 dB	
	94 dB	0.73 dB	
	104 dB	0.62 dB	
	125 Hz to 2 kHz		
	114 dB	0.35 dB	
Sound Level – Measure ¹	4 kHz		General Radio 1986 Sound Level Calibrator with Sound Level Meter
	114 dB	0.61 dB	
	125 Hz to 4 kHz		
	74 dB	0.74 dB	
	84 dB	0.63 dB	
	94 dB	0.74 dB	
	104 dB	0.63 dB	
	125 Hz to 2 kHz		
	114 dB	0.37 dB	
	4 kHz		
	114 dB	0.62 dB	

Chemical Quantities

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
pH Meters ¹	4 pH 7 pH 10 pH	0.013 pH 0.012 pH 0.012 pH	Accredited Buffer Solutions
Conductivity Meters ¹	1 µS/cm 10 µS/cm 100 µS/cm 1 000 µS/cm 10 000 µS/cm 100 000 µS/cm	0.63 µS/cm 0.63 µS/cm 2.1 µS/cm 4.8 µS/cm 43 µS/cm 370 µS/cm	Accredited Conductivity Solutions

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Capacitance – Source ¹ (Simulation)	(220 to 400) pF 10 Hz to 10 kHz	0.4 % of reading + 7.8 pF	Fluke 5522A Multiproduct Calibrator
	(0.4 to 1.1) nF 10 Hz to 10 kHz	0.4 % of reading + 7.8 pF	
	(1.1 to 3.3) nF 10 Hz to 3 kHz	0.4 % of reading + 7.8 pF	
	(3.3 to 11) nF 10 Hz to 3 kHz	0.21 % of reading + 7.8 pF	
	(11 to 33) nF 10 Hz to 1 kHz	0.2 % of reading + 78 pF	
	(33 to 110) nF 10 Hz to 1 kHz	0.21 % of reading + 78 pF	
	(110 to 330) nF 10 Hz to 1 kHz	0.2 % of reading + 0.2 nF	
	(0.33 to 1.1) µF (10 to 600) Hz	0.21 % of reading + 0.8 nF	
	(1.1 to 3.3) µF (10 to 300) Hz	0.21 % of reading + 2.3 nF	
	(3.3 to 11) µF (10 to 150) Hz	0.2 % of reading + 7.8 nF	
	(11 to 33) µF (10 to 120) Hz	0.32 % of reading + 23 nF	
	(33 to 110) µF (10 to 80) Hz	0.37 % of reading + 78 nF	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Capacitance – Source ¹ (Simulation)	(110 to 330) μ F DC to 50 Hz (0.33 to 1.1) mF DC to 20 Hz (1.1 to 3.3) mF DC to 6 Hz (3.3 to 11) mF DC to 2 Hz (11 to 33) mF DC to 0.6 Hz (33 to 110) mF DC to 0.2 Hz	0.38 % of reading + 0.2 μ F 0.35 % of reading + 0.8 μ F 0.35 % of reading + 2.3 μ F 0.35 % of reading + 7.8 μ F 0.58 % of reading + 23 μ F 0.85 % of reading + 78 μ F	Fluke 5522A Multiproduct Calibrator
Capacitance – Measure ¹	0.1 pF 100 kHz 1 pF 10 kHz 100 kHz 10 pF 1 kHz 10 kHz 100 kHz 100 pF 100 Hz 1 kHz 10 kHz 100 kHz 1 nF 20 Hz 100 Hz 1 kHz 10 kHz 100 kHz 10 nF 20 Hz 100 Hz 1 kHz 10 kHz 100 kHz	1.4 % of reading 1.4 % of reading 0.37 % of reading 1.4 % of reading 0.28 % of reading 0.28 % of reading 2.1 % of reading 0.23 % of reading 0.18 % of reading 0.21 % of reading 1.8 % of reading 0.3 % of reading 0.1 % of reading 0.1 % of reading 0.1 % of reading 0.31 % of reading 0.12 % of reading 0.09 % of reading 0.09 % of reading 0.09 % of reading	Agilent E4980A Precision LCR Meter

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Capacitance – Measure ¹	100 nF		Agilent E4980A Precision LCR Meter
	20 Hz	0.16 % of reading	
	100 Hz	0.09 % of reading	
	1 kHz	0.09 % of reading	
	10 kHz	0.09 % of reading	
	100 kHz	0.18 % of reading	
	1 μ F		
	20 Hz	0.15 % of reading	
	100 Hz	0.09 % of reading	
	1 kHz	0.09 % of reading	
	10 kHz	0.18 % of reading	
	100 kHz	0.25 % of reading	
	10 μ F		
	20 Hz	0.15 % of reading	
	100 Hz	0.09 % of reading	
	1 kHz	0.16 % of reading	
	10 kHz	0.28 % of reading	
	100 kHz	0.73 % of reading	
	100 μ F		
	20 Hz	0.16 % of reading	
	100 Hz	0.17 % of reading	
	1 kHz	0.29 % of reading	
	10 kHz	0.8 % of reading	
AC Current – Source ¹	Up to 220 μ A		Fluke 5720A Multiproduct Calibrator
	(10 to 20) Hz	0.03 % of reading + 16 nA	
	(20 to 40) Hz	0.02 % of reading + 10 nA	
	40 Hz to 1 kHz	0.02 % of reading + 8 nA	
	(1 to 5) kHz	0.03 % of reading + 12 nA	
	(5 to 10) kHz	0.11 % of reading + 65 nA	
	(0.22 to 2.2) mA		
	(10 to 20) Hz	0.03 % of reading + 40 nA	
	(20 to 40) Hz	0.02 % of reading + 35 nA	
	40 Hz to 1 kHz	0.01 % of reading + 35 nA	
	(1 to 5) kHz	0.02 % of reading + 0.1 μ A	
	(5 to 10) kHz	0.11 % of reading + 0.6 μ A	
	(2.2 to 22) mA		
	(10 to 20) Hz	0.04 % of reading + 0.4 μ A	
	(20 to 40) Hz	0.02 % of reading + 0.3 μ A	
	40 Hz to 1 kHz	0.01 % of reading + 0.3 μ A	
	(1 to 5) kHz	0.02 % of reading + 0.5 μ A	
	(5 to 10) kHz	0.11 % of reading + 5 μ A	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Current – Source ¹	(22 to 220) mA (10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (0.22 to 2.2) A 10 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.03 % of reading + 4 μ A 0.02 % of reading + 3.5 μ A 0.01 % of reading + 2.5 μ A 0.02 % of reading + 3.5 μ A 0.11 % of reading + 10 μ A 0.03 % of reading + 35 μ A 0.05 % of reading + 80 μ A 0.7 % of reading + 0.2 mA	Fluke 5720A Multiproduct Calibrator
	(2.2 to 11) A 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.05 % of reading + 0.17 mA 0.1 % of reading + 0.38 mA 0.4 % of reading + 0.75 mA	Fluke 5720A Multiproduct Calibrator, Fluke 5725A Amplifier
	(11 to 20) A (45 to 100) Hz 100 Hz to 1 kHz (1 to 5) kHz	0.09 % of reading + 4 mA 0.12 % of reading + 4 mA 2.3 % of reading + 4 mA	Fluke 5522A Multiproduct Calibrator
	(20.5 to 40) A (10 to 100) Hz 100 Hz to 1 kHz (1 to 5) kHz	0.14 % of reading + 11 mA 0.17 % of reading + 11 mA 2.3 % of reading + 11 mA	(2) Fluke 5520A Multiproduct Calibrators in Parallel
AC Current – Source ¹ Extended Frequency Ranges	(29 to 330) μ A (10 to 30) kHz (0.33 to 3.3) mA (10 to 30) kHz (3.3 to 33) mA (10 to 30) kHz (33 to 330) mA (10 to 30) kHz	1.2 % of reading + 0.3 μ A 0.78 % of reading + 0.5 μ A 0.31 % of reading + 3.1 μ A 0.31 % of reading + 0.2 mA	Fluke 5522A Multiproduct Calibrator
AC Clamp-on Ammeters (Toroidal Type) Transformer Type Sensor ¹	(20 to 150) A (45 to 65) Hz (65 to 440) Hz (150 to 1 000) A (45 to 65) Hz (65 to 440) Hz	0.3 % of reading + 26 mA 0.83 % of reading + 47 mA 0.35 % of reading + 0.1 A 1.1 % of reading + 0.2 A	Fluke 5522A Multiproduct Calibrator, 50-turn Coil

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Clamp-on Ammeters (Non-Toroidal Type) Hall Effect Sensor ¹	(45 to 150) A (45 to 65) Hz (65 to 440) Hz (150 to 1 000) A (45 to 65) Hz (65 to 440) Hz	0.57 % of reading + 0.2 A 1 % of reading + 0.2 A 0.6 % of reading + 0.9 A 1.3 % of reading + 0.9 A	
AC Current – Measure ¹	Up to 100 μ A (10 to 20) Hz (20 to 45) Hz 45 Hz to 5 kHz	0.46 % of reading + 35 nA 0.17 % of reading + 35 nA 0.07 % of reading + 35 nA	Agilent 3458A 8.5 Digit Multimeter
AC Current – Measure ¹	(0.1 to 1) mA (10 to 20) Hz (20 to 45) Hz (45 to 100) Hz 100 Hz to 5 kHz (1 to 10) mA (10 to 20) Hz (20 to 45) Hz (45 to 100) Hz 100 Hz to 5 kHz (10 to 100) mA (10 to 20) Hz (20 to 45) Hz (45 to 100) Hz 100 Hz to 5 kHz 100 mA to 1 A (10 to 20) Hz (20 to 45) Hz (45 to 100) Hz 100 Hz to 5 kHz	0.46 % of reading + 0.2 μ A 0.17 % of reading + 0.2 μ A 0.07 % of reading + 0.2 μ A 0.04 % of reading + 0.2 μ A 0.46 % of reading + 2.3 μ A 0.17 % of reading + 2.3 μ A 0.07 % of reading + 2.3 μ A 0.04 % of reading + 2.3 μ A 0.46 % of reading + 23 μ A 0.17 % of reading + 23 μ A 0.07 % of reading + 23 μ A 0.04 % of reading + 23 μ A 0.46 % of reading + 0.2 mA 0.19 % of reading + 0.2 mA 0.1 % of reading + 0.2 mA 0.12 % of reading + 0.2 mA	Agilent 3458A 8.5 Digit Multimeter
DC Current – Source ¹	200 nA to 220 μ A 220 μ A to 2.2 mA (2.2 to 22) mA (22 to 220) mA 220 mA to 2.2 A	0.004 % of reading + 6 nA 0.004 % of reading + 7 nA 0.003 % of reading + 40 nA 0.005 % of reading + 0.7 μ A 0.008 % of reading + 12 μ A	Fluke 5720A Multiproduct Calibrator
	(2.2 to 11) A (11 to 20.5) A	0.036 % of reading + 0.5 mA 0.093 % of reading + 0.6 mA	Fluke 5720A Multiproduct Calibrator, Fluke 5725A Amplifier

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
DC Clamp-on Ammeter (Non-Toroidal Type) Hall Effect Sensor ¹	(20 to 150) A (150 to 1 000) A	0.51 % of reading + 0.14 A 0.51 % of reading + 0.5 A	Fluke 5520A Multiproduct Calibrator, Wavetek Coil
DC Current – Measure ¹	Up to 100 μ A 100 μ A to 1 mA (1 to 10) mA (10 to 100) mA 100 mA to 1 A	0.003 % of reading + 0.9 nA 0.003 % of reading + 5.8 nA 0.003 % of reading + 58 nA 0.005 % of reading + 0.6 μ A 0.013 % of reading + 12 μ A	Agilent 3458A 8.5 Digit Multimeter
DC Current – Measure ¹	1 μ A to 100 mA 100 mA to 1 A (1 to 10) A (10 to 100) A (100 to 300) A	0.013 % of reading 0.015 % of reading 0.015 % of reading 0.042 % of reading 0.058 % of reading	Guideline 9211A Current Shunt, Agilent 3458A 8.5 Digit Multimeter
	(50 to 667) A	0.12 % of reading	Empro LAB-1000-100 Current Shunt, Agilent 3458A 8.5 Digit Multimeter
Inductance – Measure ¹	1 μ H 10 kHz 100 kHz 10 μ H 10 kHz 100 kHz 100 μ H 1 kHz 10 kHz 100 kHz 1 mH 100 Hz 1 kHz 10 kHz 100 kHz 10 mH 20 Hz 100 Hz 1 kHz 10 kHz 100 kHz	1.6 % of reading 0.36 % of reading 0.37 % of reading 0.2 % of reading 0.41 % of reading 0.2 % of reading 0.12 % of reading 0.56 % of reading 0.19 % of reading 0.12 % of reading 0.1 % of reading 0.86 % of reading 0.22 % of reading 0.1 % of reading 0.1 % of reading 0.11 % of reading	Agilent E4980A Precision LCR Meter

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Inductance – Measure ^{1,6}	100 mH		Agilent E4980A Precision LCR Meter
	20 Hz	0.28 % of reading	
	100 Hz	0.11 % of reading	
	1 kHz	0.1 % of reading	
	10 kHz	0.1 % of reading	
	100 kHz	0.21 % of reading	
	1 H		
	20 Hz	0.17 % of reading	
	100 Hz	0.1 % of reading	
	1 kHz	0.1 % of reading	
	10 kHz	0.11 % of reading	
	100 kHz	0.31 % of reading	
	10 H		
	20 Hz	0.15 % of reading	
	100 Hz	0.1 % of reading	
	1 kHz	0.11 % of reading	
	10 kHz	0.21 % of reading	
	100 kHz	0.69 % of reading	
	100 H		
	20 Hz	0.15 % of reading	
	100 Hz	0.11 % of reading	
	1 kHz	0.15 % of reading	
	10 kHz	0.62 % of reading	
Impedance – Measure ¹	0.1 Ω		Agilent E4980A Precision LCR Meter
	1 kHz	2 % of reading	
	10 kHz	1.2 % of reading	
	100 kHz	1.1 % of reading	
	1 Ω		
	20 Hz	0.67 % of reading	
	100 Hz	0.45 % of reading	
	1 kHz	0.36 % of reading	
	10 kHz	0.33 % of reading	
	100 kHz	0.31 % of reading	
	10 Ω		
	20 Hz	0.29 % of reading	
	100 Hz	0.2 % of reading	
	1 kHz	0.17 % of reading	
	10 kHz	0.18 % of reading	
	100 kHz	0.18 % of reading	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Impedance – Measure ¹	100 Ω		Agilent E4980A Precision LCR Meter
	20 Hz	0.16 % of reading	
	100 Hz	0.09 % of reading	
	1 kHz	0.09 % of reading	
	10 kHz	0.12 % of reading	
	100 kHz	0.12 % of reading	
	1 k Ω		
	20 Hz	0.15 % of reading	
	100 Hz	0.09 % of reading	
	1 kHz	0.09 % of reading	
	10 kHz	0.09 % of reading	
	100 kHz	0.09 % of reading	
	10 k Ω		
	20 Hz	0.15 % of reading	
	100 Hz	0.09 % of reading	
	1 kHz	0.09 % of reading	
	10 kHz	0.09 % of reading	
	100 kHz	0.1 % of reading	
	100 k Ω		
	20 Hz	0.17 % of reading	
	100 Hz	0.1 % of reading	
	1 kHz	0.1 % of reading	
	10 kHz	0.17 % of reading	
	100 kHz	0.28 % of reading	
DC Resistance – Source/Measure ¹	Up to 10 Ω	0.002 % of reading + 58 $\mu\Omega$	Agilent 3458A 8.5 Digit Multimeter, Decade Resistor
	(10 to 100) Ω	0.001 % of reading + 0.58 m Ω	
	100 Ω to 1 k Ω	0.001 % of reading + 0.58 m Ω	
	(1 to 10) k Ω	0.001 % of reading + 5.8 m Ω	
	(10 to 100) k Ω	0.001 % of reading + 58 m Ω	
	100 k Ω to 1 M Ω	0.002 % of reading + 2.3 Ω	
	(1 to 10) M Ω	0.006 % of reading + 0.12 k Ω	
	(10 to 100) M Ω	0.06 % of reading + 1.2 k Ω	
DC Resistance – Source ¹ (Fixed Artifact)	100 M Ω to 1 G Ω	0.6 % of reading + 12 k Ω	Empro LAB-500-5 Current Shunt IET Labs DCCS/0.0001 Current Shunt Fluke 742A-1 Standard Resistor Fluke 742A-10k Standard Resistor
	10 $\mu\Omega$	30 n Ω	
	100 $\mu\Omega$	0.13 $\mu\Omega$	
	1 Ω	10 $\mu\Omega$	
	10 k Ω	52 m Ω	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
DC Resistance – Source ¹ (Fixed Artifact)	333.33 $\mu\Omega$ 1 m Ω 10 m Ω 100 m Ω 1 Ω 10 Ω 100 Ω 1 k Ω 10 k Ω	0.19 $\mu\Omega$ 0.42 $\mu\Omega$ 1.7 $\mu\Omega$ 17 $\mu\Omega$ 0.15 m Ω 1.5 m Ω 15 m Ω 0.15 Ω 1.5 Ω	Guideline 9211A Multi-tap Current Shunt
DC Resistance – Source ¹ (Variable Artifact)	100 k Ω to 1 M Ω (1 to 10) M Ω (10 to 100) M Ω 100 M Ω to 1 G Ω (1 to 10) G Ω (10 to 100) G Ω 100 G Ω to 1 T Ω	0.035 % of reading 0.035 % of reading 0.12 % of reading 0.3 % of reading 0.58 % of reading 1.2 % of reading 2.6 % of reading	High Resistance Box (Up to 5 kV)
DC Resistance – Source ¹ (Simulation)	Up to 11 Ω (11 to 33) Ω (33 to 111) Ω (110 to 330) Ω 330 Ω to 1.1k Ω (1.1 to 3.3) k Ω (3.3 to 11) k Ω (11 to 33) k Ω (33 to 110) k Ω (110 to 330) k Ω 330 k Ω to 1.19 M Ω (1.1 to 3.3) M Ω (3.3 to 11) M Ω (11 to 33) M Ω (33 to 110) M Ω (110 to 330) M Ω 330 M Ω to 1.1 G Ω	32 $\mu\Omega/\Omega$ + 0.8 m Ω 24 $\mu\Omega/\Omega$ + 1.2 m Ω 22 $\mu\Omega/\Omega$ + 1.1 m Ω 22 $\mu\Omega/\Omega$ + 1.6 m Ω 22 $\mu\Omega/\Omega$ + 1.6 m Ω 22 $\mu\Omega/\Omega$ + 1.6 m Ω 22 $\mu\Omega/\Omega$ + 1.6 m Ω 22 $\mu\Omega/\Omega$ + 0.2 Ω 22 $\mu\Omega/\Omega$ + 0.2 Ω 27 $\mu\Omega/\Omega$ + 1.6 Ω 26 $\mu\Omega/\Omega$ + 1.6 Ω 66 $\mu\Omega/\Omega$ + 23 Ω 0.1 m Ω/Ω + 39 Ω 0.2 m Ω/Ω + 1.9 k Ω 0.4 m Ω/Ω + 2.3 k Ω 0.23 % of reading + 78 k Ω 12 % of reading + 0.4 M Ω	Fluke 5522A Multiproduct Calibrator
AC Voltage – Source ¹	Up to 2.2 mV (10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.16 % of reading + 4 μ V 0.1 % of reading + 4 μ V 0.08 % of reading + 4 μ V 0.13 % of reading + 4 μ V 0.17 % of reading + 5 μ V 0.33 % of reading + 10 μ V 0.47 % of reading + 20 μ V 0.58 % of reading + 20 μ V	Fluke 5720A Multiproduct Calibrator

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage – Source ¹	(2.2 to 22) mV		Fluke 5720A Multiproduct Calibrator
	(10 to 20) Hz	0.04 % of reading + 4 μ V	
	(20 to 40) Hz	0.03 % of reading + 4 μ V	
	40 Hz to 20 kHz	0.01 % of reading + 4 μ V	
	(20 to 50) kHz	0.03 % of reading + 4 μ V	
	(50 to 100) kHz	0.06 % of reading + 5 μ V	
	(100 to 300) kHz	0.12 % of reading + 10 μ V	
	(300 to 500) kHz	0.16 % of reading + 20 μ V	
	500 kHz to 1 MHz	0.27 % of reading + 20 μ V	
	(22 to 220) mV		
	(10 to 20) Hz	0.03 % of reading + 12 μ V	
	(20 to 40) Hz	0.01 % of reading + 7 μ V	
	40 Hz to 20 kHz	0.009 % of reading + 7 μ V	
	(20 to 50) kHz	0.02 % of reading + 7 μ V	
	(50 to 100) kHz	0.05 % of reading + 17 μ V	
	(100 to 300) kHz	0.09 % of reading + 20 μ V	
	(300 to 500) kHz	0.14 % of reading + 25 μ V	
	500 kHz to 1 MHz	0.28 % of reading + 45 μ V	
	(0.22 to 2.2) V		
	(10 to 20) Hz	0.03 % of reading + 40 μ V	
	(20 to 40) Hz	0.01 % of reading + 15 μ V	
	40 Hz to 20 kHz	0.005 % of reading + 8 μ V	
	(20 to 50) kHz	0.008 % of reading + 10 μ V	
	(50 to 100) kHz	0.01 % of reading + 30 μ V	
	(100 to 300) kHz	0.04 % of reading + 80 μ V	
	(300 to 500) kHz	0.1 % of reading + 0.2 mV	
	500 kHz to 1 MHz	0.18 % of reading + 0.3 mV	
	(2.2 to 22) V		
	(10 to 20) Hz	0.03 % of reading + 0.4 mV	
	(20 to 40) Hz	0.01 % of reading + 0.1 mV	
	40 Hz to 20 kHz	0.005 % of reading + 50 μ V	
	(20 to 50) kHz	0.008 % of reading + 0.1 mV	
	(50 to 100) kHz	0.011 % of reading + 0.2 mV	
	(100 to 300) kHz	0.03 % of reading + 0.6 mV	
	(300 to 500) kHz	0.1 % of reading + 2 mV	
	500 kHz to 1 MHz	0.17 % of reading + 3.2 mV	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage – Source ¹	(22 to 220) V (10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz (220 to 1 100) V 40 Hz to 1 kHz (1 to 20) kHz (20 to 30) kHz	0.03 % of reading + 4 mV 0.01 % of reading + 1.5 mV 0.006 % of reading + 0.6 mV 0.009 % of reading + 1 mV 0.02 % of reading + 2.5 mV 0.09 % of reading + 16 mV 0.44 % of reading + 40 mV 0.8 % of reading + 80 mV 0.01 % of reading + 4 mV 0.02 % of reading + 6 mV 0.06 % of reading + 11 mV	Fluke 5720A Multiproduct Calibrator
	(220 to 750) V (30 to 50) kHz (50 to 100) kHz	0.06% of reading + 11 mV 0.23 % of reading + 45 mV	Fluke 5720A Multiproduct Calibrator, Fluke 5725A Amplifier
AC Voltage – Measure ¹	Up to 1 mV 100 kHz to 1 MHz (1 to 3) MHz (3 to 10) MHz (10 to 20) MHz (1 to 3) mV 100 kHz to 1 MHz (1 to 3) MHz (3 to 10) MHz (10 to 20) MHz (3 to 100) mV 100 kHz to 1 MHz (1 to 3) MHz (3 to 10) MHz (10 to 20) MHz (20 to 30) MHz	1.8 % of reading + 2.4 μ V 3.5 % of reading + 2.4 μ V 9.3 % of reading + 2.4 μ V 23 % of reading + 2.4 μ V 1 % of reading + 2 μ V 3.5 % of reading + 2 μ V 9.3 % of reading + 2 μ V 23 % of reading + 2 μ V 0.9 % of reading + 3 μ V 1.8 % of reading + 3 μ V 2.9 % of reading + 3 μ V 7 % of reading + 3 μ V 14 % of reading + 3 μ V	Rohde & Schwarz URE3 RMS Voltmeter

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage – Measure ¹	Up to 10 mV		Agilent 3458A Opt 002 8.5 Digit Multimeter
	(1 to 40) Hz	0.04 % of reading + 3.5 μ V	
	40 Hz to 1 kHz	0.04 % of reading + 1.2 μ V	
	(1 to 20) kHz	0.04 % of reading + 1.2 μ V	
	(20 to 50) kHz	0.15 % of reading + 1.2 μ V	
	(50 to 100) kHz	0.6 % of reading + 1.2 μ V	
	(100 to 300) kHz	4.6 % of reading + 2.3 μ V	
	300 kHz to 1 MHz	1.5 % of reading + 5.8 μ V	
	(1 to 4) MHz	8 % of reading + 8.1 μ V	
	(10 to 100) mV		
	(1 to 40) Hz	0.01 % of reading + 4.6 μ V	
	40 Hz to 1 kHz	0.01 % of reading + 2.3 μ V	
	(1 to 20) kHz	0.02 % of reading + 2.3 μ V	
	(20 to 50) kHz	0.04 % of reading + 2.3 μ V	
	(50 to 100) kHz	0.09 % of reading + 2.3 μ V	
	(100 to 300) kHz	0.36 % of reading + 12 μ V	
	300 kHz to 1 MHz	1.2 % of reading + 12 μ V	
	(1 to 2) MHz	1.9 % of reading + 12 μ V	
	(2 to 4) MHz	4.7 % of reading + 81 μ V	
	(4 to 8) MHz	4.7 % of reading + 92 μ V	
	(8 to 10) MHz	17 % of reading + 0.12 mV	
	(0.1 to 1) V		
	(1 to 40) Hz	0.01 % of reading + 46 μ V	
	40 Hz to 1 kHz	0.01 % of reading + 23 μ V	
	(1 to 20) kHz	0.02 % of reading + 23 μ V	
	(20 to 50) kHz	0.04 % of reading + 23 μ V	
	(50 to 100) kHz	0.09 % of reading + 23 μ V	
	(100 to 300) kHz	0.35 % of reading + 0.12 mV	
	300 kHz to 1 MHz	1.2 % of reading + 0.12 mV	
	(1 to 2) MHz	1.9 % of reading + 0.12 mV	
	(2 to 4) MHz	4.6 % of reading + 0.81 mV	
	(4 to 8) MHz	4.6 % of reading + 0.92 mV	
	(8 to 10) MHz	17 % of reading + 1.2 mV	
	(1 to 10) V		
	(1 to 40) Hz	0.01 % of reading + 0.46 mV	
	40 Hz to 1 kHz	0.01 % of reading + 0.23 mV	
	(1 to 20) kHz	0.02 % of reading + 0.23 mV	
	(20 to 50) kHz	0.04 % of reading + 0.23 mV	
	(50 to 100) kHz	0.09 % of reading + 0.23 mV	
	(100 to 300) kHz	0.35 % of reading + 1.2 mV	
	300 kHz to 1 MHz	1.2 % of reading + 1.2 mV	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage – Measure ¹	(1 to 10) V (1 to 2) MHz (2 to 4) MHz (4 to 8) MHz (8 to 10) MHz	1.8 % of reading + 1.2 mV 4.6 % of reading + 8.1 mV 4.7 % of reading + 9.2 mV 17 % of reading + 12 mV	Agilent 3458A Opt 002 8.5 Digit Multimeter
	(10 to 100) V (1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz (100 to 700) V (1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.02 % of reading + 4.6 mV 0.02 % of reading + 2.3 mV 0.02 % of reading + 2.3 mV 0.04 % of reading + 2.3 mV 0.14 % of reading + 2.3 mV 0.46 % of reading + 12 mV 1.7 % of reading + 12 mV 0.05 % of reading + 46 mV 0.05 % of reading + 23 mV 0.07 % of reading + 23 mV 0.14 % of reading + 23 mV 0.35 % of reading + 23 mV	
AC High Voltage – Measure ¹	(0.7 to 10) kV (20 to 100) Hz (100 to 400) Hz	0.14 % of reading + 0.4 V 0.48 % of reading + 0.2 V	Vitretek 4700 Digital HV Meter, Associated High Voltage Probes
	(10 to 30) kV (30 to 70) Hz (70 to 200) Hz (200 to 450) Hz (30 to 50) kV (30 to 70) Hz (70 to 200) Hz (200 to 450) Hz (50 to 70) kV (30 to 70) Hz (70 to 200) Hz	0.11 % of reading + 2.4 V 0.7 % of reading + 2.4 V 1.4 % of reading + 2.4 V 0.13 % of reading + 2.5 V 0.7 % of reading + 2.5 V 2.9 % of reading + 2.5 V 0.16 % of reading + 2.6 V 1.2 % of reading + 2.6 V	
DC Voltage – Source ¹	Up to 220 mV (0.22 to 2.2) V (2.2 to 11) V (11 to 22) V (22 to 220) V	0.000 9 % of reading + 0.4 μ V 0.000 5 % of reading + 0.7 μ V 0.000 4 % of reading + 2.5 μ V 0.000 4 % of reading + 4 μ V 0.000 6 % of reading + 40 μ V	Fluke 5720A Multiproduct Calibrator
	(220 to 1 100) V	0.000 8 % of reading + 0.4 mV	Fluke 5720A Multiproduct Calibrator, Fluke 5725A Amplifier

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
DC Voltage – Measure ¹	Up to 100 mV (0.1 to 1) V (1 to 10) V (10 to 100) V (100 to 500) V (500 to 800) V (800 to 1 000) V	0.000 8 % of reading + 0.6 μ V 0.000 7 % of reading + 0.6 μ V 0.000 5 % of reading + 0.6 μ V 0.000 8 % of reading + 35 μ V 0.001 % of reading + 0.1 mV 0.002 % of reading + 0.1 mV 0.002 % of reading + 0.1 mV	Agilent 3458A Opt 002 8.5 Digit Multimeter
DC High Voltage – Measure ¹	(1 to 10) kV (10 to 20) kV (20 to 70) kV (70 to 100) kV	0.04 % of reading + 92 mV 0.09 % of reading + 2.4 V 0.09 % of reading + 2.4 V 0.17 % of reading + 2.5 V	Vitrek 4700 Digital HV Meter, Associated High Voltage Probes
AC Power – Source ^{1,3} PF = 1	(3.3 to 9) mA (10 to 65) Hz (0.11 mW to 3) mW 3 mW to 9 W (9 to 33) mA (10 to 65) Hz (0.3 to 10) mW 10 mW to 33 W (33 to 90) mA (10 to 65) Hz (1 to 30) mW 30 mW to 90 W (90 to 330) mA (10 to 65) Hz (3 to 100) mW 100 mW to 300 W (0.33 to 0.9) A (10 to 65) Hz (11 to 300) mW (0.3 to 900) W (0.9 to 2.2) A (10 to 65) Hz (30 to 720) mW 0.72 W to 2 kW (2.2 to 4.5) A (10 to 65) Hz 80 mW to 1.4 W 1.4 W to 4.5 kW (4.5 to 20.5) A (10 to 65) Hz 150 mW to 20 kW	0.13 % of reading 0.08 % of reading 0.09 % of reading 0.08 % of reading 0.07 % of reading 0.06 % of reading 0.09 % of reading 0.08 % of reading 0.07 % of reading 0.08 % of reading 0.09 % of reading 0.08 % of reading 0.09 % of reading 0.18 % of reading 0.17 % of reading	Fluke 5522A Multiproduct Calibrator

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
DC Power – Source ¹ (0.33 to 330) mA (0.33 to 3) A (3 to 20.5) A	11 μ W to 1.1 mW (1.1 to 110) mW 110 mW to 110 W (110 to 330) W 11 μ W to 110 mW 110 mW to 990 W 990 W to 3 kW 99 mW to 0.99 W 0.99 W to 6.8 kW (6.8 to 20.5) kW	0.02 % of reading 0.03 % of reading 0.02 % of reading 0.02 % of reading 0.04 % of reading 0.05 % of reading 0.01 % of reading 0.09 % of reading 0.07 % of reading 0.04 % of reading	Fluke 5522A Multiproduct Calibrator
Low Frequency Phase – Source ¹	Up to 179.99 ° (10 to 65) Hz (65 to 500) Hz 500 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.1 ° 0.2 ° 0.4 ° 1.9 ° 3.9 ° 7.8 °	Fluke 5522A Multiproduct Calibrator
Oscilloscopes ¹ Amplitude – DC into 50 Ω load into 1 M Ω load Amplitude – Square Wave into 50 Ω load into 1 M Ω load Time Markers into 50 Ω load	(-6 to 6) V (-130 to 130) V 10 Hz to 100 kHz 1 mV p-p to 6.6 Vp-p 10 Hz to 1 kHz 1 mV p-p to 6.6 Vp-p (1 kHz to 10) kHz 1 mV p-p to 6.6 Vp-p 1 ns to 20 ms 50 ms 0.1 s 0.2 s 0.5 s	0.2 % of reading + 31 μ V 0.04 % of reading + 31 μ V 0.19 % of reading + 31 μ V 0.08 % of reading + 31 μ V 0.19 % of reading + 31 μ V 0.000 2 % of reading 2.3 μ s 7.6 μ s 28 μ s 0.2 ms	Fluke 5520A/11 Multiproduct Calibrator

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Oscilloscopes ^{1,4}			Fluke 5520A/11 Multiproduct Calibrator
Time Markers into 50 Ω load	1 s 2 s 5 s	0.6 ms 2.4 ms 15 ms	
Rise Time into 50 Ω load	5 mVp-p to 2.5 Vp-p		
Rate: 1 kHz to 2 MHz	250 ps (nominal)	51 ps	
Rate: 2 MHz to 10 MHz	250 ps (nominal)	51 ps	
Leveled Sine Wave into 50 Ω load	5 mVp-p to 5 Vp-p 50 kHz 50 kHz to 100 MHz (100 to 300) MHz (300 to 600) MHz (600 to 1 100) MHz	1.8 % of reading + 0.2 mV 2.8 % of reading + 0.2 mV 3.2 % of reading + 0.2 mV 4 % of reading + 0.2 mV 5.5 % of reading + 0.2 mV	
Bandwidth/Flatness (50 kHz Reference) into 50 Ω load	5 mVp-p to 5.5 Vp-p 50 kHz to 100 MHz (100 to 300) MHz (300 to 600) MHz (600 to 1 100) MHz	1.4 % of reading + 78 μ V 1.8 % of reading + 78 μ V 3.2 % of reading + 78 μ V 3.9 % of reading + 78 μ V	
Input Impedance – Measure	(40 to 60) Ω (0.5 to 1.5) M Ω	0.082 % of reading 0.081 % of reading	
Input Capacitance – Measure	(5 to 50) pF	3.9 % of reading + 0.4 pF	
Waveform Generator (Sine, Square, Triangle) Amplitude	10 Hz to 10 kHz		
into 50 Ω load	1.8 mVp-p to 2.5 Vp-p	2.3 % of reading + 78 μ V	
into 1 M Ω load	1.8 mVp-p to 55 Vp-p	2.3 % of reading + 78 μ V	
Frequency	10 Hz to 10 kHz	0.002 % of reading + 12 mHz	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Electrical Simulation of Thermocouple Indicating Devices – Measure/Source ¹	Type B		Ectron 1140A Thermocouple Calibrator/Simulator
	(250 to 350) °C	1.2 °C	
	(350 to 445) °C	0.9 °C	
	(445 to 580) °C	0.71 °C	
	(580 to 750) °C	0.55 °C	
	(750 to 1 000) °C	0.45 °C	
	(1 000 to 1 820) °C	0.35 °C	
	Type C		
	(0 to 250) °C	0.24 °C	
	(250 to 1 000) °C	0.19 °C	
	(1 000 to 1 500) °C	0.21 °C	
	(1 500 to 1 800) °C	0.24 °C	
	(1 800 to 2 000) °C	0.27 °C	
	(2 000 to 2 250) °C	0.33 °C	
	(2 250 to 2 315) °C	0.37 °C	
	Type D		
	(0 to 100) °C	0.34 °C	
	(100 to 300) °C	0.28 °C	
	(300 to 1 400) °C	0.18 °C	
	(1 400 to 1 650) °C	0.19 °C	
	(1 650 to 1 930) °C	0.23 °C	
	(1 930 to 2 100) °C	0.28 °C	
	(2 100 to 2 200) °C	0.3 °C	
	(2 200 to 2 320) °C	0.34 °C	
	Type E		
	(-270 to -245) °C	1.6 °C	
	(-245 to -195) °C	0.24 °C	
	(-195 to -155) °C	0.12 °C	
	(-155 to -90) °C	0.09 °C	
	(-90 to 0) °C	0.08 °C	
	(0 to 15) °C	0.08 °C	
	(15 to 890) °C	0.06 °C	
	(890 to 1 000) °C	0.07 °C	
	Type G		
	(0 to 100) °C	1.6 °C	
	(100 to 300) °C	0.5 °C	
	(300 to 600) °C	0.35 °C	
	(600 to 1 760) °C	0.18 °C	
	(1 760 to 2 030) °C	0.2 °C	
	(2 030 to 2 200) °C	0.25 °C	
	(2 200 to 2 315) °C	0.27 °C	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Electrical Simulation of Thermocouple Indicating Devices – Measure/Source ¹	Type J		Ectron 1140A Thermocouple Calibrator/Simulator
	(-210 to -180) °C	0.15 °C	
	(-180 to -120) °C	0.12 °C	
	(-120 to -50) °C	0.09 °C	
	(-50 to 990) °C	0.08 °C	
	(990 to 1 200) °C	0.09 °C	
	Type K		
	(-270 to -255) °C	2.5 °C	
	(-255 to -195) °C	0.85 °C	
	(-195 to -115) °C	0.16 °C	
	(-115 to -55) °C	0.12 °C	
	(-55 to 1 000) °C	0.09 °C	
	(1 000 to 1 372) °C	0.1 °C	
	Type N		
	(-270 to -260) °C	5.4 °C	
	(-260 to -200) °C	1.5 °C	
	(-200 to -140) °C	0.29 °C	
	(-140 to -70) °C	0.18 °C	
	(-70 to 25) °C	0.14 °C	
	(25 to 160) °C	0.12 °C	
	(160 to 1 300) °C	0.11 °C	
	Type PLII		
	(0 to 100) °C	0.1 °C	
	(100 to 925) °C	0.08 °C	
	(925 to 1 200) °C	0.1 °C	
	(1 200 to 1 395) °C	0.11 °C	
	Type R		
	(-50 to -30) °C	0.8 °C	
	(-30 to 45) °C	0.69 °C	
	(45 to 160) °C	0.49 °C	
	(160 to 380) °C	0.35 °C	
	(380 to 775) °C	0.3 °C	
	(775 to 1 768) °C	0.26 °C	
	Type S		
	(-50 to -30) °C	0.76 °C	
	(-30 to 45) °C	0.68 °C	
	(45 to 105) °C	0.49 °C	
	(105 to 310) °C	0.41 °C	
	(310 to 615) °C	0.35 °C	
	(615 to 1 768) °C	0.31 °C	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Electrical Simulation of Thermocouple Indicating Devices – Measure/Source ¹	Type T (-270 to -255) °C (-255 to -240) °C (-240 to -210) °C (-210 to -150) °C (-180 to -40) °C (-40 to 100) °C (100 to 400) °C	1.9 °C 0.6 °C 0.36 °C 0.22 °C 0.15 °C 0.09 °C 0.08 °C	Ectron 1140A Thermocouple Calibrator/Simulator
Electrical Simulation of RTD Indicators – Source ¹	Pt 385, 100 Ω (-200 to -80) °C (-80 to 0) °C (0 to 100) °C (100 to 300) °C (300 to 400) °C (400 to 630) °C (630 to 800) °C Pt 385, 200 Ω (-200 to -80) °C (-80 to 0) °C (0 to 100) °C (100 to 260) °C (260 to 300) °C (300 to 400) °C (400 to 600) °C (600 to 630) °C Pt 385, 500 Ω (-200 to -80) °C (-80 to 0) °C (0 to 100) °C (100 to 260) °C (260 to 300) °C (300 to 400) °C (400 to 600) °C (600 to 630) °C	0.05 °C 0.05 °C 0.07 °C 0.09 °C 0.1 °C 0.12 °C 0.23 °C 0.04 °C 0.04 °C 0.04 °C 0.05 °C 0.12 °C 0.13 °C 0.14 °C 0.16 °C 0.04 °C 0.05 °C 0.05 °C 0.06 °C 0.08 °C 0.08 °C 0.09 °C 0.11 °C	Fluke 5520A Multiproduct Calibrator

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Electrical Simulation of RTD Indicators – Source ¹	Pt 385, 1 000 Ω		Fluke 5520A Multiproduct Calibrator
	(-200 to -80) °C	0.03 °C	
	(-80 to 0) °C	0.03 °C	
	(0 to 100) °C	0.04 °C	
	(100 to 260) °C	0.05 °C	
	(260 to 300) °C	0.06 °C	
	(300 to 400) °C	0.07 °C	
	(400 to 600) °C	0.07 °C	
	(600 to 630) °C	0.23 °C	
	Pt 3916, 100 Ω		
	(-200 to -190) °C	0.25 °C	
	(-190 to -80) °C	0.04 °C	
	(-80 to 0) °C	0.05 °C	
	(0 to 100) °C	0.06 °C	
	(100 to 260) °C	0.07 °C	
	(260 to 300) °C	0.08 °C	
	(300 to 400) °C	0.09 °C	
	(400 to 600) °C	0.1 °C	
	(600 to 630) °C	0.23 °C	
	Pt 3926, 100 Ω		
	(-200 to -80) °C	0.05 °C	
	(-80 to 0) °C	0.05 °C	
	(0 to 100) °C	0.07 °C	
	(100 to 300) °C	0.09 °C	
	(300 to 400) °C	0.1 °C	
	(400 to 630) °C	0.12 °C	
	PtNi 385, 120 Ω		
	(-80 to 0) °C	0.08 °C	
	(0 to 100) °C	0.08 °C	
	(100 to 260) °C	0.14 °C	
	Cu 427, 10 Ω		
	(-100 to 260) °C	0.3 °C	

Electrical – RF/Microwave

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Relative RF Power	100 kHz to 4.2 GHz (-30 to -20) dB (-20 to -10) dB (-10 to 0) dB	1.9 % of reading 1.8 % of reading 1.8 % of reading	Agilent 437B Power Meter, Agilent 8482A Power Sensor

Electrical – RF/Microwave

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Relative RF Power	100 kHz to 4.2 GHz (0 to 10) dB (10 to 20) dB	1.8 % of reading 1.8 % of reading	Agilent 437B Power Meter, Agilent 8482A Power Sensor

Length – Dimensional Metrology

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Gage Blocks ² Length	Up to 1 in (1 to 4) in (5 to 20) in Up to 100 mm (100 to 500) mm	2.5 µin (0.5 + 1.9L) µin (1.5 + 1.3L) µin (0.013 + 0.002L) µm or 64 nm, whichever is greater (0.16 + 0.001L) µm	Gage Block Comparator, Master Gage Blocks
Parallelism	Up to 1 inDL (25.4 mm)	1 µin (25 nm)	Optical Parallels
Flatness	Up to 1 inDL (25.4 mm)	3 µin (76 nm)	Optical Flats
Angle Measuring Devices ^{1,2}	5 ° (5 to 20) ° (20 to 35) ° (35 to 45) ° (45 to 60) ° (60 to 75) ° (75 to 85) ° 90 °	1.7 " 2.4 " 3.6 " 4.6 " 8.1 " 17 " 52 " 1.1 "	Sine Bar, Gage Blocks, Surface Plate, Granite Square
Calipers, Micrometers ^{1,2} Travel (Outside, Inside, Depth, Step)	Up to 0.5 in (0.5 to 1) in (1 to 4) in (4 to 15) in (15 to 40) in	13 µin 14 µin (10 + 3.6L) µin (11 + 4.2L) µin (11 + 4.4L) µin	B89.1 Grade 0 Gage Blocks
Anvil Flatness	Up to 1 inDL	6.2 µin	Optical Flats
Anvil Parallelism	Up to 1 inDL	6.6 µin	Optical Parallels

Length – Dimensional Metrology

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Bore Gages ^{1,2}	(0.125 to 0.25) in (0.25 to 1) in (1 to 6) in	34 µin 36 µin (30 + 7L) µin	Characterized Ring Gages
Indicators ^{1,2} (Dial, Digital, Drop, Test)	Up to 6 in	(6.3 + 3.2L) µin	Gage Blocks, Surface Plate
Test Indicators	Up to 0.025 in	5.6 µin	Universal Length Measuring Machine
Wire Crimpers/Dies ¹ Die Diameter	(0.011 to 0.625) in	53 µin	Pin Gages
Crimp Height	Up to 0.8 in	180 µin	Micrometer
Height Gages ^{1,2}	Up to 4 in (4 to 24) in	(26 + 0.5L) µin (16 + 3L) µin	Gage Blocks
Length – Single Axis ² Outside Dimension	Up to 1 in (1 to 7) in (7 to 12) in	(6 + 1L) µin (4 + 3.5 L) µin (4L) µin	Universal Length Measuring Machine
Inside Dimension	(0.04 to 1) in (1 to 2.5) in (2.5 to 10) in (10 to 14) in	11 µin 17 µin (18 + 3L) µin (38 + 3L) µin	
Surface Plates ^{1,2}			In accordance with ASME B89.3.7 using Wyler Levels
Overall Flatness	Up to 250 inDL	(38 + 1√DL) µin	Gage Amp, Probe Level Plate, Surface Plate
Local Area Flatness (Repeat Readings)			
Up to 12.37 inDL	Up to 0.001 in	58 µin	
(12.37 to 18.25) inDL	Up to 0.001 in	65 µin	
(18.97 to 30.59) inDL	Up to 0.001 in	65 µin	Repeat-o-Meter
> 16.97 inDL	Up to 0.001 in	33 µin	
Optical Comparators ^{1,2} X-Y Length	Up to 6 in	(100 + 14L) µin	Calibration Grids
Squareness	(0.04 to 1) in	(120 + 1.5L) µin	Calibration Grids
Magnification	10X to 50X	(240 + 21L) µin	Magnification Checker
Optical Reference Plane ¹	Up to 6 in (6 to 12) in	51 µin 75 µin	Glass Scale, Calibration Grids

Length – Dimensional Metrology

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Laser Micrometers ^{1,2}	Up to 0.1 in (0.1 to 0.4) in (0.4 to 1) in	13 µin 8 µin (11 + 5L) µin	Characterized Master Pin Gages
Surface Roughness Testers ¹	Ra Rmax	70 nm (3 µin) 0.91 µm (36 µin)	Calibrated Specimen
Micrometer Head Spindle Displacement	Up to 1 in Up to 25.4 mm	10 µin 0.26 µm	Heidenhain Linear Encoder
Micrometer Standards ²	Up to 36 in (36 to 80) in Up to 900 mm (900 to 2 000) mm	(30 + 4L) µin (200 + 5L) µin (0.8 + 0.004L) µm (5 + 0.005L) µm	P&W Measuring Machine
Pin Gages ¹ (Non-contact)	(0.004 to 1) in	33 µin	Laser Micrometer
Plug Gages ² (Outside Diameter)	Up to 1 in (1 to 7) in	12 µin (3 + 3L) µin	Universal Length Measuring Machine
Measuring Rules ²	Up to 48 in	(30 + 8L) µin	CSIP with Heidenhain LIP 401R Linear Encoder
Thread Wires	(2 to 120) TPI (0.008 33 to 0.5) in (2 to 120) TPI (0.211 6 to 12.7) mm	12 µin 0.3 µm	Universal Length Measuring Machine
Cylindrical Ring Gages ² Inside Diameter	(0.04 to 1) in (1 to 2.5) in (2.5 to 10) in (10 to 14) in	11 µin 17 µin (18 + 3L) µin (38 + 3L) µin	Universal Length Measuring Machine
Thread Plug Gages ² Pitch Diameter, 60 ° Thread	Up to 1 in (1 to 4) in (4 to 7) in	79 µin 80 µin 83 µin	Universal Length Measuring Machine, Thread Wires
Major Diameter	Up to 1 in (1 to 7) in	13 µin (10 + 3L) µin	Universal Length Measuring Machine
Step Height	Up to 1 in	32 µin	Gage Amp, Probe, Gage Blocks

Length – Dimensional Metrology

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Thread Ring Gages Inner Pitch Diameter	Up to 1 in (1 to 4) in (4 to 7) in	79 μ in 80 μ in 83 μ in	Master Thread Plug Uncertainty
Tapered Thread Plug Gages Pitch Diameter Taper	Up to 3 in	90 μ in	Universal Length Measuring Machine, Thread Wires
Standoff	Up to 1 in	31 μ in	Gage Amp, Probe, Gage Blocks
Tapered Thread Ring Gage	Up to 3 in	90 μ in	Master Tapered Thread Plug Uncertainty
Thickness Foils, Feeler Gages	Up to 1 in	(6 + 1L) μ in	Universal Length Measuring Machine

Mass and Mass Related

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Balances, Scales ^{1,5} Metric (SI)	Up to 500 mg 500 mg to 5 g (5 to 10) g (10 to 30) g (30 to 100) g 100 g to 5 kg (5 to 10) kg	12 μ g 40 μ g 61 μ g 94 μ g 0.000 9 % of reading 0.000 3 % of reading 0.000 3 % of reading	ASTM E617 Class 1 Weights and internal calibration procedure utilized for the calibration of the weighing system.
	(5 to 50) mg (50 to 500) mg 500 mg to 5 g (5 to 10) g (10 to 20) g 20 g to 2 kg	16 μ g 29 μ g 63 μ g 88 μ g 0.12 mg 0.006 % of reading	ASTM E617 Class 2 Weights and internal calibration procedure utilized for the calibration of the weighing system.
Balances, Scales ^{1,5} Metric (SI)	450 g to 27 kg	0.014 % of reading	ASTM E617 Class 6 Weights and internal calibration procedure utilized for the calibration of the weighing system.
Avoirdupois	(1 to 62) lb	0.014 % of reading	

Mass and Mass Related

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Balances, Scales ^{1,5} Metric (SI) Avoirdupois	(450 to 211) kg (1 to 466) lb	0.012 % of reading 0.012 % of reading	NIST Class F Weights and internal calibration procedure utilized for the calibration of the weighing system.
Rockwell Hardness and Superficial Testers ^{1,7}	HRA Scale (45 to 55) HRA (70 to 80) HRA (80 to 88) HRA HRBw Scale (44 to 49) HRBw (73 to 79) HRBw (88 to 99) HRBw HRC Scale (24 to 28) HRC (42 to 47) HRC (60 to 65) HRC HREw Scale (69 to 76) HREw (83 to 91) HREw (96 to 100) HREw HR15TW Scale (72 to 75) HR15TW (82 to 86) HR15TW (90 to 92) HR15TW HR30N Scale (45 to 48) HR30N (65 to 68) HR30N (76 to 78) HR30N HR30TW Scale (48 to 53) HR30TW (59 to 63) HR30TW (72 to 81) HR30TW HR45TW Scale (25 to 30) HR45TW (43 to 49) HR45TW (61 to 64) HR45TW	0.6 HRA 0.37 HRA 0.29 HRA 0.84 HRBw 0.75 HRBw 0.64 HRBw 0.44 HRC 0.42 HRC 0.38 HRC 0.62 HREw 0.63 HREw 0.59 HREw 0.64 HR15TW 0.64 HR15TW 0.45 HR15TW 0.63 HR30N 0.65 HR30N 0.51 HR30N 0.67 HR30TW 0.55 HR30TW 0.48 HR30TW 0.65 HR45TW 0.65 HR45TW 0.6 HR45TW	Indirect verification per ASTM E18 using hardness test blocks.
Brinell Hardness Testers ^{1,7}	HBW Scale Low Medium High	1.4 HBW 4.2 HBW 5.3 HBW	Indirect verification per ASTM E10 using hardness test blocks.

Mass and Mass Related

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Knoop Hardness Testers ^{1,7}	HK 0.1 Scale (858 to 886) HK	20 HK	Indirect verification per ASTM E384, ASTM E92 using hardness test blocks.
Leeb Hardness Testers ^{1,7}	HLD Scale (752 to 759) HLD (786 to 789) HLD	9.9 HLD 9.4 HLD	Indirect verification per ASTM A596 using hardness test blocks.
Vickers Hardness Testers ^{1,7}	HV 0.3 Scale (476 to 494) HV (747 to 757) HV HV 0.5 Scale (493 to 519) HV HV 1 Scale (460 to 468) HV (741 to 752) HV	10 HV 12 HV 12 HV 8 HV 13 HV	Indirect verification per ASTM E384, ASTM E92 using hardness test blocks.
Absolute Pressure ¹ (Pneumatic)	Up to 30 psia (14.7 to 45) psia (14.7 to 65) psia (14.7 to 115) psia (14.7 to 165) psia (14.7 to 265) psia (14.7 to 315) psia (14.7 to 515) psia (14.7 to 765) psia (14.7 to 1 515) psia	0.003 6 psi 0.005 3 psi 0.007 psi 0.012 psi 0.018 psi 0.031 psi 0.037 psi 0.062 psi 0.09 psi 0.19 psi	Mensor APC6000 Pressure Controller
	(14.7 to 507.5) psia (507.5 to 1 015) psia (14.7 to 1 507.5) psia (1 507.5 to 3 015) psia (14.7 to 3 007.5) psia (3 007.5 to 6 015) psia	0.05 psi 0.01 % of reading + 0.001 5 psi 0.15 psi 0.01 % of reading + 0.001 5 psi 0.3 psi 0.01 % of reading + 0.001 5 psi	Mensor CPC8000 Pressure Controller
Gage Pressure ¹ (Pneumatic)	Up to 0.5 psig Up to 1 psig Up to 5 psig Up to 10 psig Up to 15 psig Up to 30 psig Up to 50 psig Up to 100 psig Up to 150 psig	0.000 06 psi 0.000 1 psi 0.000 6 psi 0.001 2 psi 0.001 8 psi 0.003 6 psi 0.006 psi 0.012 psi 0.018 psi	Mensor APC6000 Pressure Controller

Mass and Mass Related

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Gage Pressure ¹ (Pneumatic)	Up to 250 psig Up to 300 psig Up to 500 psig Up to 750 psig Up to 1 500 psig	0.031 psi 0.037 psi 0.062 psi 0.093 psi 0.19 psi	Mensor APC6000 Pressure Controller
	Up to 500 psig (500 to 1 000) psig Up to 1 500) psig (1 500 to 3 000) psig Up to 3 000) psig (3 000 to 6 000) psig	0.05 psi 0.01 % of reading 0.15 psi 0.01 % of reading 0.3 psi 0.01 % of reading	Mensor CPC8000 Pressure Controller
Gage Pressure ¹ (Hydraulic)	Up to 10 000 psig	1.2 psi	Comparison to Mensor 2106 Digital Pressure Gage
	(100 to 16 000) psig	0.025 % of reading	Budenberg 580HXA Pressure Balance
Torque Drivers, Indicators ¹	(10 to 100) ozf·in (4 to 50) lbf·in (30 to 400) lbf·in	1.2 % of reading 1.1 % of reading 1.1 % of reading	CDI Torque Transducers
Torque Wrenches ¹	(4 to 250) lbf·ft (60 to 600) lbf·ft	0.54 % of reading 0.49 % of reading	
Torque Analyzers, Torque Transducers	(2 to 40) ozf·in 5 ozf·in to 25 lbf·in (2.5 to 150) lbf·ft (50 to 250) lbf·ft (25 to 1 000) lbf·ft	0.13 % of reading 0.085 % of reading 0.055 % of reading 0.053 % of reading 0.052 % of reading	NIST Class F Weights, Torque Wheels, Torque Arm

Thermodynamic

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Humidity – Measure ¹	(-40 to -20) °C (0 to 95) RH%	1.7 % of reading + 1.8 %RH	Vaisala MI70/HMP76B Temp/Humidity Indicator/Probe
	(-20 to 15) °C (0 to 95) RH% (15 to 25) °C (0 to 90) RH% (90 to 100) RH%	0.92 % of reading + 1.3 %RH 1.3 RH% 2 %RH	

Thermodynamic

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Humidity – Measure ¹	(25 to 40) °C (0 to 95) RH% (40 to 180) °C (0 to 95) RH%	0.92 % of reading + 1.3 %RH 1.7 % of reading + 1.8 %RH	Vaisala MI70/HMP76B Temp/Humidity Indicator/Probe
Thermo-Hygrometers ¹ Temperature	(5 to 95) %RH (18 to 28) °C	0.27 °C	Kaymont 2000 Humidity Generator, Vaisala MI70/HMP76B Temp/Humidity Indicator/Probe
Humidity	(18 to 25) °C (7 to 90) %RH (90 to 100) %RH	1.4 %RH 2.1 %RH	
Temperature – Measure ¹	(-195 to 0) °C (0 to 420) °C (420 to 660) °C	0.012 °C 0.026 °C 0.034 °C	SPRT, Temperature Indicator
	(600 to 1 093) °C (1 093 to 1 200) °C	0.47 % of reading + 0.9 °C 0.47 % of reading + 1.9 °C	Type N T/C Probe, Temperature Indicator
	(600 to 1 093) °C (1 093 to 1 200) °C	0.47 % of reading + 0.9 °C 0.47 % of reading + 1.9 °C	Type K T/C Probe, Temperature Indicator
	(615 to 1 093) °C (1 093 to 1 200) °C	0.15 % of reading + 1 °C 0.15 % of reading + 1.9 °C	Type S T/C Probe, Temperature Indicator
Temperature – Source	0.01	0.83 Mk	Triple Point Water Cell
Temperature – Source ¹	(-30 to -20) °C (-20 to 0) °C (0 to 150) °C (150 to 200) °C	0.043 °C 0.018 °C 0.03 °C 0.05 °C	Fluke 7320 Bath, Fluke 6102 Bath, SPRT, Temperature Indicator
	(200 to 420) °C (420 to 660) °C	0.26 °C 0.41 °C	Fluke 9173 Dry-well, SPRT, Temperature Indicator
Infrared Thermometers ¹	(-15 to 0) °C (0 to 50) °C (50 to 100) °C (100 to 120) °C (120 to 200) °C (200 to 350) °C (350 to 500) °C	0.8 °C 0.6 °C 0.7 °C 0.8 °C 1 °C 1.6 °C 2 °C	Blackbody Source (Flat Plate) $\epsilon = (0.9 \text{ to } 1)$ $\lambda = (8 \text{ to } 14) \mu\text{m}$

Time and Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Reference Frequency (Source)	10 MHz	640 pHz/Hz	Rubidium Oscillator
Frequency – Measure ¹	10 MHz	2.1 μ Hz/Hz	HP 53132A Opt 10 Universal Frequency Counter
Stopwatches, Timers ¹	Up to 599 s/mon	58 ms/day	Vibrograf 4500 Timometer

DIMENSIONAL MEASUREMENT

1 Dimensional

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Dimension Measurement – 1D	X-axis Up to 1 in (1 to 3) in (3 to 6) in Y-axis Up to 2 in (2 to 3) in (3 to 5) in	210 μ in 370 μ in 480 μ in 360 μ in 410 μ in 560 μ in	Optical Comparator used as a Reference Standard for Dimensional Inspection. (Length)
Dimensional Measurement – 1D ^{1,2}	Up to 12 in	(28 + 3.2L) μ in	Gage Amp, Probe, and Gage Blocks utilized as Reference Standards for Dimensional Inspection. (Height)
	Up to 4 inDL Up to 12.37 inDL (12.37 to 18.25) inDL (18.97 to 30.59) inDL	6.2 μ in 35 μ in 42 μ in 46 μ in	Optical Flats, Gage Amp, and Indicator utilized as Reference Standards for Dimensional Inspection. (Flatness)
	Up to 12.37 inDL (12.37 to 18.25) inDL (18.97 to 30.59) inDL	33 μ in 40 μ in 45 μ in	Gage Amp and Indicator utilized as Reference Standards for Dimensional Inspection. (Parallelism)

1 Dimensional

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Dimensional Measurement – 1D ^{1,2}	Up to 18 in	163 µin	Gage Amp, Surface Plate, and Granite Square utilized as Reference Standards for Dimensional Inspection. (Squareness)
	Up to 12 in (12 to 18) in (18 to 30) in	35 µin 42 µin 46 µin	Gage Amp and Indicator utilized as Reference Standards for Dimensional Inspection. (Straightness)

2 Dimensional

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Angle ²	Up to 180 °	28.8"	Optical Comparator utilized as Reference Standard for Dimensional Inspection.
Radius	Up to 1 in (1 to 3) in (3 to 6) in	260 µin 450 µin 590 µin	Optical Comparator utilized as Reference Standard for Dimensional Inspection.

3 Dimensional

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Dimensional Measurement – 3D ²	X = Up to 2 500 mm Y = Up to 1 500 mm Z = Up to 1 000 mm	(6 + 0.009L) µm	Coordinate Measurement Machine utilized as Reference Standard for Dimensional Measurement.
	X = Up to 98.4 in Y = Up to 59 in Z = Up to 39.3 in	(236 + 0.4L) µin	

Calibration and Measurement Capability (CMC) is expressed in terms of the measurement parameter, measurement range, expanded uncertainty of measurement and reference standard, method, and/or equipment. The expanded uncertainty of measurement is expressed as the standard uncertainty of the measurement multiplied by a coverage factor of 2 ($k=2$), corresponding to a confidence level of approximately 95%.

Notes:

1. On-site calibration service is available for this parameter, since on-site conditions are typically more variable than those in the laboratory, larger measurement uncertainties are expected on-site than what is reported on the accredited scope.
2. L = length in measurand unit (inches or millimeters); " = arc-second; DL = diagonal length.
3. The uncertainties shown are for the most favorable conditions. There is an increase in uncertainty that corresponds to the laboratory's AC voltage and current uncertainties at different frequencies other than the ones shown. Power factors (PF) other than the one shown contribute to the power uncertainty. PF is related to the cosine of phase. Therefore, uncertainties track the laboratory's phase uncertainty closely at PF near one, but are magnified heavily as PF approaches zero. The lab may also report reactive power, apparent power, and power factor under this accreditation. If needed, contact laboratory for more information regarding uncertainties at frequency and power factor combinations other than the ones shown.
4. The stated uncertainty is the laboratory's ability to source a fast rise pulse that is approximately 250 ps. In the typical application of measuring rise time of an oscilloscope, this value is one of the contributing factors, but other factors are derived from the DUT.
5. The CMC for scales and balances is highly dependent upon the resolution of the unit under test. The CMC presented here does not include the resolution of the unit under test. The resolution will be included in the reported measurement uncertainty at the time of calibration.
6. As frequency & amplitude deviate from the listed values, uncertainty may be higher than stated. If needed, contact laboratory for more information regarding uncertainties at frequency and range combinations other than the ones shown.
7. The following statement is from the NIST recommended practice guide (Special Publication 960-5, page 60, section 8.3.3, paragraph 2) "Currently, there are no generally agreed upon U.S. or international methods for calculating the measurement uncertainty of a Rockwell hardness machine or the uncertainty in the certified value of standardized test blocks".
8. This scope is formatted as part of a single document including Certificate of Accreditation No. AC-2489.23.



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